

Arunabha Majumder

Mechanical Engineer | Sub-System Design & Ownership | Test Rigs | SolidWorks | Fabrication

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PROFESSIONAL SUMMARY

Mechanical Engineer with a B.Tech in Mechanical Engineering, an M.Tech in Mechatronics and about three years of professional R&D experience owning an electromechanical sub-system end to end: concept generation, architecture and requirements, SolidWorks CAD and drawings, hands-on workshop fabrication, assembly, integration, commissioning and testing. Designs test rigs to validate hypotheses quickly, builds analytical models in Python and MATLAB, and uses FEA before committing to fabrication. Reasons from first principles and questions every part: removed a force/torque sensor from a robot design, cutting cost, mass and part count. Certified SolidWorks Associate (CSWA); Best Paper Award, IFToMM ISRM 2026.

CORE SKILLS

- **System Design:** Concept generation, system architecture, technical requirements, sub-system ownership, integration
- **CAD & Design:** SolidWorks (CSWA certified), Fusion 360, design drawings, Design for Manufacture and Assembly (DFM/DFA), cost and part-count reduction
- **Fabrication & Build:** Hands-on workshop fabrication and build, machined parts, additive manufacturing / 3D printing, assembly, commissioning
- **Testing:** Test-rig design, test planning, running experiments, data acquisition (DAQ), data analysis
- **Simulation & Modelling:** Analytical modelling, Python, MATLAB/Simulink, Finite Element Analysis (FEA), stiffness and kinematic models validated against experiments
- **Electromechanical:** Pneumatic systems (valves, pressure sensors), motors, encoders, force sensors, embedded control

PROFESSIONAL EXPERIENCE

Research Engineer – Mechanical Sub-System Design & Development (PhD Project: VIEXO) | 2023 – Present

Aalborg University, Aalborg, Denmark

- Owned an actuated wearable sub-system end to end: concept generation, architecture and technical requirements, SolidWorks CAD and drawings, fabrication, assembly, integration, commissioning and test.
- Re-thought the actuation architecture from first principles, pairing a parallel spring with a 6 Nm motor to offload gravity torque; cut energy consumption by more than 25% (Best Paper Award, IFToMM ISRM 2026).
- Questioned the need for an external force/torque sensor in a 5-bar parallel robot and replaced it with force estimation from the actuator's torque–deflection characteristic, reducing cost, mass, part count and integration complexity.
- Designed custom automated test rigs to validate hypotheses before fabrication; analytical stiffness models matched experimental hardware to below 10% RMS error.
- Built prototypes hands-on from machined and additively manufactured parts and integrated motor, encoder, force sensors, DAQ and embedded control.
- Wrote and executed test plans, including a 12-participant study across 3 assistance modes and 2 load conditions, with an end-to-end data acquisition and analysis pipeline.
- Taught university students and supervised Bachelor's (B.Tech) students (2023 – 2025).

M.Tech Researcher – Pneumatic Actuator Systems (Master's Thesis) | 2020 – 2022

CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

- Fabricated pneumatic artificial muscle actuators in-house from raw materials and built the test setup with valves, pressure sensors and DAQ to characterise load–displacement behaviour.

- Derived an empirical actuator model from test data and designed a neural-network-based gain-scheduled PID controller, reducing position-tracking error to 0.3–0.78%.

Mechanical Engineering Capstone – Upper-Limb Rehabilitation Exoskeleton | 2018 – 2019
Siddaganga Institute of Technology, Tumakuru, India

- Led CAD design, fabrication and user testing end to end; won the Best Major Project Award and an INR 100,000 TATA Technologies Innovation Grant.

HANDS-ON BUILD PROJECTS

- **Go-Kart (2017):** Led end-to-end design, fabrication and build of a custom Go-Kart from the ground up.
- **Efficycle (2018):** Worked in a multidisciplinary team to design and build a human-electric hybrid vehicle for the Efficycle competition.

EDUCATION

Ph.D. Candidate, Mechatronics & Robotics | 2023 – Present
Aalborg University, Department of Materials and Production, Aalborg, Denmark

M.Tech, Mechatronics | 2020 – 2022
AcSIR – CSIR-Central Mechanical Engineering Research Institute (CMERI), Durgapur, India

B.Tech, Mechanical Engineering | 2015 – 2019
Siddaganga Institute of Technology, Visvesvaraya Technological University (VTU), Tumakuru, India

CERTIFICATIONS & AWARDS

- Certified SolidWorks Associate (CSWA) – Mechanical Design
- Best Research Paper Award, 9th IFToMM International Symposium on Robotics and Mechatronics (2026)
- Best Major Project Award, Department of Mechanical Engineering (2019)
- TATA Technologies Innovation Award – INR 100,000 Grant (2019)

PUBLICATIONS

- Majumder, A., Wagner, J.W., Zhu, Y., Oliveira, A.S., and Bai, S. "A Hybrid Actuated Shoulder Exoskeleton for Energy-Efficient Upper Arm Support." *Robotics and Mechatronics: ISRM 2026, MMS*, vol. 213, Springer.
- Majumder, A., Sarkar, D., Chakraborty, S., Singh, A., Roy, S.S., and Arora, A. "Neural Network-Based Gain Scheduled Position Control of a Pneumatic Artificial Muscle." *IEEE CONECCT*, 2022.

TECHNICAL TOOLS

- **CAD & Manufacturing:** SolidWorks, Fusion 360, FEA, Bambu Lab and Ultimaker Cura (3D printing)
- **Software & Modelling:** Python, MATLAB/Simulink, LabVIEW (basic), Arduino
- **Testing & Instrumentation:** Qualisys Motion Capture, Forsentec Force/Torque Sensors, Pneumatic Control, Custom Rig Fabrication

LANGUAGES

English (C1, fluent) | Bengali (Native) | Hindi (Conversational)

REFERENCES

Available upon request.